

PRACTICAL NO. 7

Aim

A Case Study on “Technology Life Cycle” of a Successful Entrepreneur

1. Introduction

Technology Life Cycle represents the different stages through which a technology develops, becomes widely used, reaches maturity and may eventually be replaced by newer technology.

Understanding the Technology Life Cycle helps an entrepreneur identify technological changes and make suitable decisions regarding adoption, improvement and replacement of technology.

For this case study, Elon Musk and the development and adoption of electric vehicle technology through Tesla are considered.

2. About the Entrepreneur – Elon Musk

Elon Musk is an entrepreneur and the CEO of Tesla. He has played an important role in promoting electric vehicle technology and sustainable transportation.

He focused on combining technology, performance and sustainability. Tesla helped demonstrate how electric vehicles could be developed as practical and desirable products for consumers.

3. Technology Selected

Electric Vehicle Technology

The selected technology for this case study is electric vehicle technology.

Electric vehicles use electric motors, rechargeable batteries, electronic control systems and charging infrastructure. Tesla integrated these technologies into modern electric vehicles and helped increase the adoption of electric mobility.

4. Technology Life Cycle

The Technology Life Cycle can be represented through the following stages:

Introduction → Growth → Maturity → Decline / Replacement → New or Improved Technology

Stage 1 – Introduction

In the introduction stage, a new or improved technology is introduced to the market.

Tesla introduced electric vehicles as a modern alternative to conventional petrol and diesel vehicles. At this stage, consumers were becoming familiar with battery-powered transportation and charging requirements.

Entrepreneur's role:

- Identify a technological opportunity.
- Understand customer requirements.

- Introduce the technology in a useful product.
- Create awareness among customers.

Stage 2 – Growth

In the growth stage, the technology starts gaining wider acceptance and its use increases.

Electric vehicles became increasingly popular as battery technology, vehicle range, charging systems and supporting infrastructure improved. Many automobile companies started developing electric vehicles.

Entrepreneur's role:

- Improve the technology and product.
- Respond to customer feedback.
- Introduce new features.
- Adapt to increasing competition.

Stage 3 – Maturity

In the maturity stage, the technology becomes widely accepted and established in the market.

Electric vehicles are becoming an important part of modern transportation. Companies focus on improving battery performance, charging speed, safety, vehicle efficiency and software features.

Entrepreneur's role:

- Continue product improvement.
- Maintain quality.
- Introduce useful innovations.
- Keep the product relevant to changing customer needs.

Stage 4 – Decline / Replacement

Technology may enter a decline stage when newer and more advanced technologies begin to replace older technologies.

For electric vehicles, individual technologies such as older battery chemistries, charging methods and electronic components may gradually be replaced by improved technologies. The complete electric vehicle technology does not necessarily disappear; instead, its components and methods continue to evolve.

Entrepreneur's role:

- Identify outdated technology.
- Adopt newer technologies.
- Upgrade existing products.

- Prepare for future technological changes.

5. Technology Life Cycle Diagram

TECHNOLOGY LIFE CYCLE

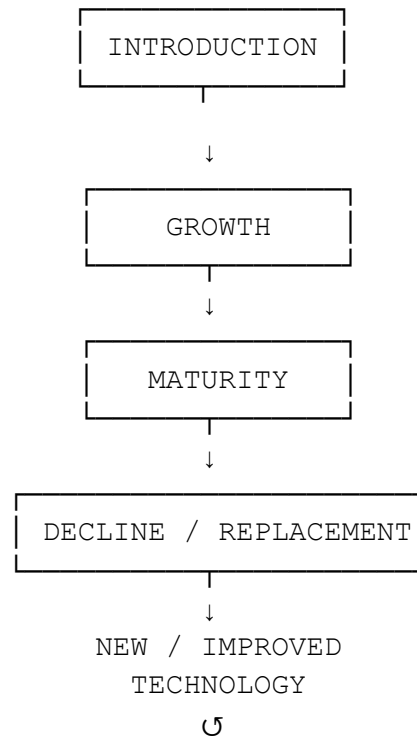


Fig.: Technology Life Cycle

6. Interpretation of the Case Study

The case of Elon Musk and Tesla shows how an entrepreneur can use technological changes to develop innovative products.

Tesla combined battery technology, electric motors, software and charging systems into a practical consumer product. As technology developed, Tesla continued to improve its vehicles and adopt newer technologies.

This shows that an entrepreneur should not depend on one technology permanently. The entrepreneur should observe technological changes, understand customer requirements and adopt suitable technologies at the appropriate stage of the Technology Life Cycle.

7. Role of Technology Life Cycle in Entrepreneurship

Technology Life Cycle helps an entrepreneur to:

1. Identify new technological opportunities.
2. Understand the stage of a technology.
3. Decide when to adopt or improve technology.
4. Respond to changing customer requirements.

5. Replace outdated technology.
6. Plan for future technological developments.

8. Learning from the Case Study

The following points can be learned from the case of Elon Musk:

- Innovation is important for business growth.
- Technology should be used according to customer needs.
- Entrepreneurs should continuously monitor technological changes.
- Existing products should be improved with changing technology.
- Outdated technology should be replaced at the appropriate time.
- Successful enterprises need to adapt to technological changes.

9. Conclusion

The case study of Elon Musk and Tesla explains the importance of the Technology Life Cycle in entrepreneurship.

Technology passes through different stages from introduction to growth, maturity and eventual replacement by newer or improved technology. An entrepreneur who understands these stages can make better decisions regarding technology adoption, improvement and replacement.

Thus, the Technology Life Cycle is useful for understanding how technological changes can influence the development and growth of an enterprise.